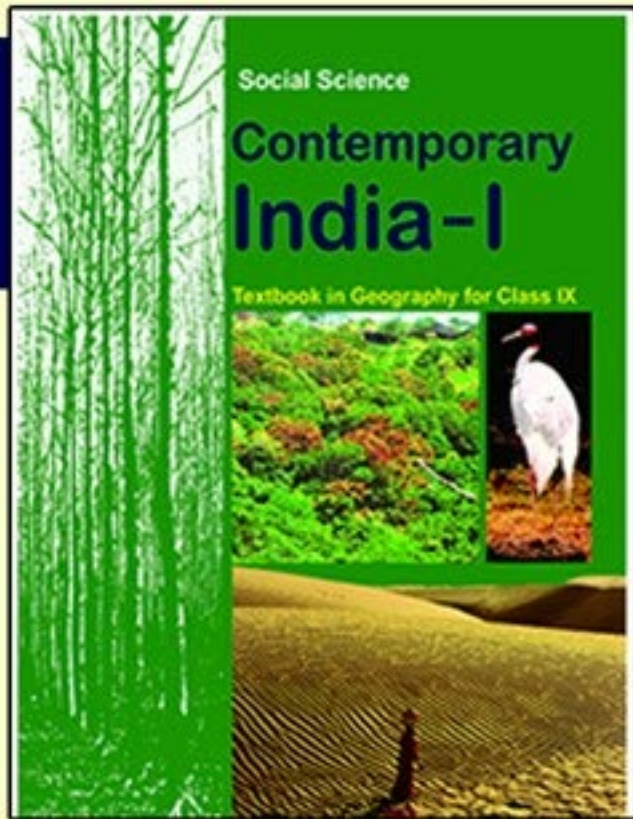




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NCERT BASICS



GEOGRAPHY

THE DRAINAGE SYSTEM OF INDIA

MODULE - 9



MAJOR DRAINAGE BASINS OF INDIA



TERMS YOU NEED TO UNDERSTAND!

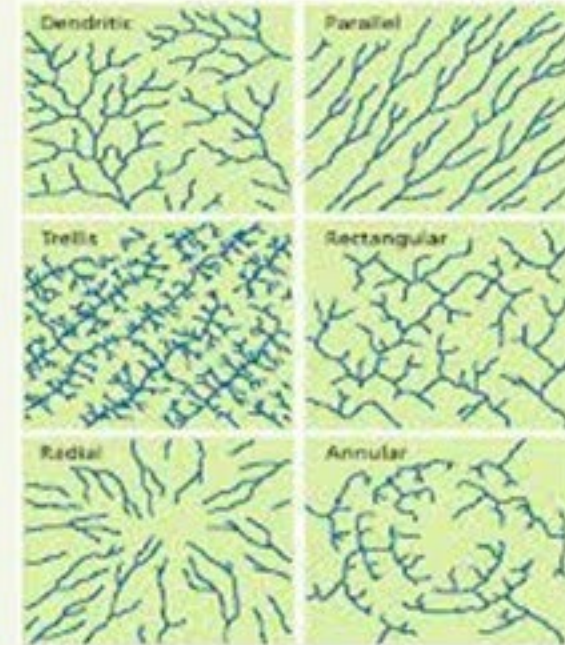
- The flow of water through well-defined channels is known as '**Drainage**'.
- The network of such channels is called a '**Drainage System**'.
- A river drains the water collected from a specific area, which is called its '**Catchment Area**'.
- An area drained by a river and its tributaries is called a '**Drainage Basin**'.





WHAT DO YOU MEAN BY DRAINAGE PATTERN?

- Geometric arrangement of streams in region gets determined by:
 - Slope
 - Rock resistance to weathering and erosion
 - Climate
 - Structural controls of the landscape
- This kind of geometric arrangement of the streams is known as a Drainage Pattern.

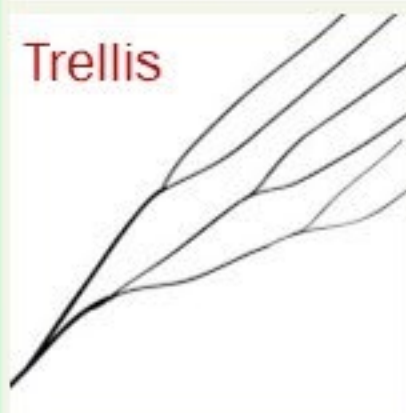


IMPORTANT DRAINAGE PATTERNS

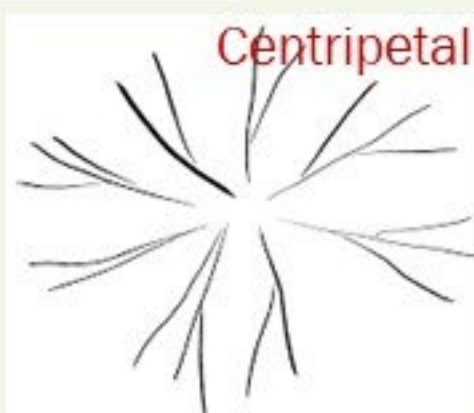
- The drainage pattern resembling the branches of a tree is known as “**Dendritic**”.
 - Ex. Brahmaputra.
- When the rivers originate from a hill and flow in all directions, the drainage pattern is known as ‘**Radial**’.
 - Ex: The rivers originating from the Amarkantak range.
- When the primary tributaries of rivers flow parallel to each other and secondary tributaries join them at right angles, the pattern is known as ‘**Trellis**’.
 - Ex: Ganga, Yamuna
- When the rivers discharge their waters from all directions in a lake or depression, the pattern is known as ‘**Centripetal**’.
 - Ex: Loktak lake in Manipur.



Dendritic drainage of the Yarlung Tsangpo River, in Tibet as seen from space



Trellis



Centripetal



DRAINAGE SYSTEM IN INDIA

MAJOR DRAINAGE BASINS OF INDIA

Indian drainage system is the outcome of the evolutionary process of the three major physiographic units and the nature and characteristics of precipitation.





CLASSIFICATION OF DRAINAGE SYSTEM IN INDIA

1. Discharge of water

Arabian Sea
Drainage

Bay of Bengal
Drainage

2. Size of Watershed

Major River Basin

Medium River Basin

Minor River Basin

Himalayan Rivers

Indus

Ganga

Brahmaputra

3. Origin, Nature & Characteristics

Peninsular Rivers

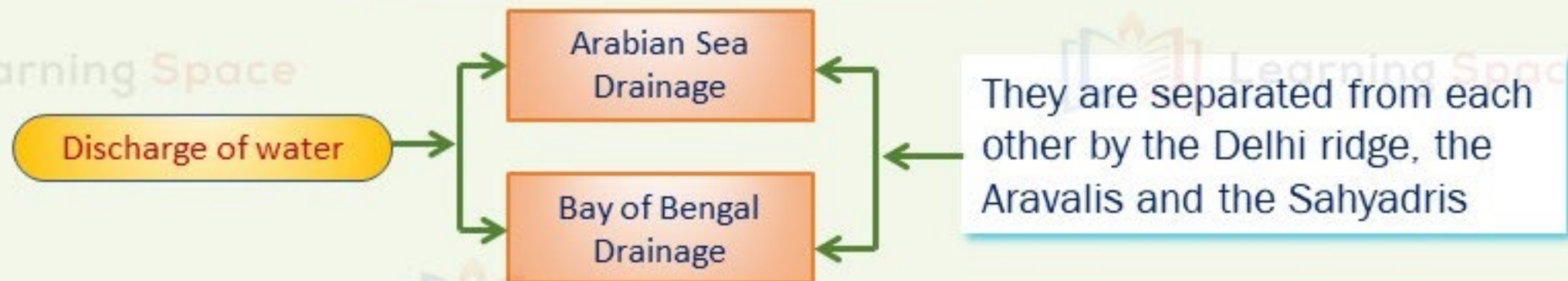
West Flowing
Rivers

East Flowing
Rivers





1. DISCHARGE OF WATER



ARABIAN SEA DRAINAGE

- The Arabian Sea drainage or Western drainage receive less rainfall.
- It orients approximately 23 per cent of the drainage area of the country.
- Indian Rivers that flow into Arabian Sea are seasonal or non-perennial

BAY OF BENGAL DRAINAGE

- Bay of Bengal Drainage or Eastern drainage experience good rainfall.
- It orients approximately 77 per cent of the drainage area of the country.
- Most of the Himalayan waters (perennial rivers) drain into Bay of Bengal.





RIVERS OF THE INLAND DRAINAGE BASIN (ENDORHEIC BASIN):

- When a river does not reach the sea but disappears into the sand, such a region is called an area of **Inland Drainage**.
- Inland drainage streams are short-lived.
- The following are few examples:
 - The **Ghaggar river** in Haryana, which is supposed to be remnant of the proverbial ancient Saraswati river, gets lost in dry sands near Hanumangarh in Rajasthan.
 - The **Luni river** in Rajasthan originates near Ajmer and after flowing through Thar desert gets lost in the Rann of Kutch.



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2. SIZE OF WATERSHED

WATERSHED: It is an area of land that drains or “sheds” water into a specific waterbody. Watersheds drain rainfall and snowmelt into streams and rivers.

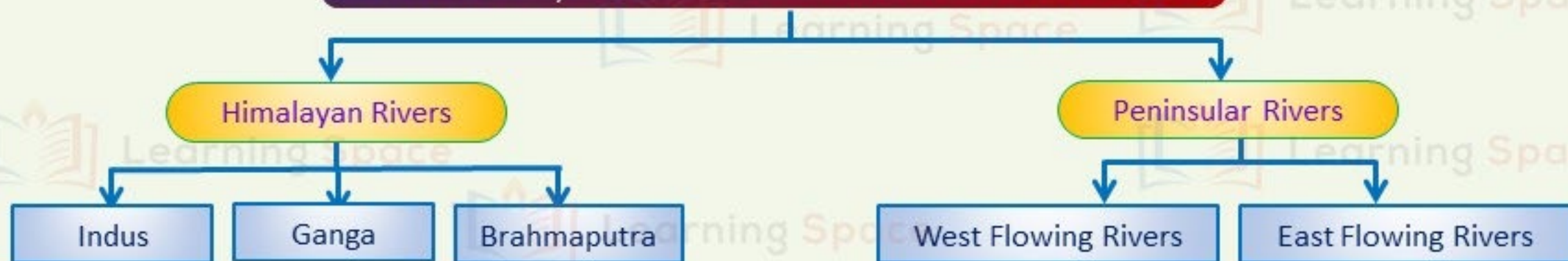


- **Major river basins** – With more than 20,000 sq. km of catchment area.
- It includes drainage basins such as the Ganga, Brahmaputra, the Krishna, Tapi, Narmada, Mahi, Penna, Sabarmati etc.

- **Medium river basins** – With catchment area between 2,000-20,000 sq. km incorporating river basins such as Kalindi, Periyar, Meghna, etc.

- **Minor river basins** – With catchment area of less than 2,000 sq. Kms.

3. ORIGIN, NATURE & CHARACTERISTICS



I. EVOLUTION OF THE HIMALAYAN DRAINAGE SYSTEM

Geologists believe that:

1. A mighty river called Shiwalik or Indo-Brahma, traversed the entire longitudinal extent of the Himalaya from Assam to Punjab and onwards to Sind, and finally discharged into the Gulf of Sind near lower Punjab during the Miocene period some 5-24 million years ago.

2. It is opined that in due course of time Indo-Brahma river was dismembered into three main drainage systems.

MAJOR DRAINAGE BASINS OF INDIA



3. ORIGIN, NATURE & CHARACTERISTICS

I. HIMALAYAN DRAINAGE SYSTEM

THE INDUS RIVER SYSTEM

- It is one of the largest river basins of the world.
- The Indus also known as the Sindhu, is the westernmost of the Himalayan rivers in India.
- It originates from a glacier near Bokhar Chu in the Kailash Mountain range. In Tibet, it is known as 'Singi Khamban'; or Lion's mouth.
- It finally discharges into the Arabian Sea, east of Karachi.
- A little over one-third of the Indus basin is located in India; in the states of Jammu & Kashmir, Himachal Pradesh and Punjab.

- ✓ The **right-bank tributaries** of Indus: Shyok, Gilgit, Hunza, Nubra, Kabul, Khurram, Tochi, Gomai, Sangar, Kunar.
- ✓ The **left-bank tributaries** of Indus: Zaskar, Satluj, Beas, Ravi, Chenab, Jhelum.



3. ORIGIN, NATURE & CHARACTERISTICS

I. HIMALAYAN DRAINAGE SYSTEM

THE INDUS RIVER SYSTEM

Chief Characteristics:

- The Indus drains the largest number of glaciers of Himalayas, Ladakh, Zaskar, and Kailash.
- The Indus plain has a very gentle slope.

DID YOU KNOW!

The waters of the Indus river system are shared by India and Pakistan according to the **Indus Water Treaty** signed between the two countries on 19th September 1960.



3. ORIGIN, NATURE & CHARACTERISTICS

I. HIMALAYAN DRAINAGE SYSTEM

THE GANGA RIVER SYSTEM



- ✓ Important **left bank tributaries** of the Ganga: Ramganga, Gomati, Ghaghara, Gandak, Kosi, Mahanand.
- ✓ Important **right bank tributaries** of the Ganga: Son, Yamuna.

- The Ganga river system is the largest in India.
- It rises in the Gangotri glacier near Gaumukh in the Uttarkashi district of Uttaranchal. Here, it is known as the Bhagirathi. At Devprayag, the Bhagirathi meets the Alaknanda; hereafter, it is known as the Ganga.
- The river finally discharges itself into the Bay of Bengal near the Sagar Island.

3. ORIGIN, NATURE & CHARACTERISTICS

I. HIMALAYAN DRAINAGE SYSTEM

THE GANGA RIVER SYSTEM



- This river is having a number of perennial and non-perennial rivers originating in the Himalayas in the north and the Peninsula in the south, respectively.
- It accounts for 26.3% of the geographical area of the country and is shared by ten states.

3. ORIGIN, NATURE & CHARACTERISTICS

I. HIMALAYAN DRAINAGE SYSTEM

THE BRAHMAPUTRA RIVER SYSTEM

- The Brahmaputra has its origin in the Chemayungdung glacier of the Kailash range near the Mansarovar lake.
- It flows eastwards parallel to the Himalayas.
- Most of the course of the Brahmaputra lies in Tibet, popularly known as Tsangpo.
- Dihang is joined by Dibang, Lohit, Kenula and many other tributaries and finally forms the Brahmaputra in Assam.



DID YOU KNOW!

Majuli is a river island in the Brahmaputra River, Assam and in 2016 it became the first island to be made a district in India.

3. ORIGIN, NATURE & CHARACTERISTICS

I. HIMALAYAN DRAINAGE SYSTEM

THE BRAHMAPUTRA RIVER SYSTEM

➤ Characteristic Features:

- River Brahmaputra is a little longer than the river Indus.
- It forms a spectacular Grand Canyon – like canyon in Tibet.
- The river is nearly 16 km wide at Dibrugarh and forms many islands, the most important one is Majuli.
- The Brahmaputra is well-known for floods, channel shifting and bank erosion. This is due to the fact that most of its tributaries are large, and bring large quantity of sediments owing to heavy rainfall in its catchment area.



Arabian Sea



Bay of Bengal

- ✓ Major **left bank tributaries**: Burhi Dihing, Dhansari (South), Kalang.
- ✓ Major **right bank tributaries**: Subansiri, Kameng, Manas, Sankosh.

3. ORIGIN, NATURE & CHARACTERISTICS



II. THE EVOLUTION OF PENINSULAR DRAINAGE SYSTEM

Three major geological events in the distant past have shaped the present drainage systems of Peninsular India.

- (i) Subsidence of the western flank of the Peninsula leading to its submergence below the sea during the early tertiary period
- (ii) Upheaval of the Himalayas when the northern flank of the Peninsular block was subjected to subsidence and the consequent trough faulting.
- (iii) Slight tilting of the Peninsular block from northwest to the southeastern direction

MAJOR DRAINAGE BASINS OF INDIA





3. ORIGIN, NATURE & CHARACTERISTICS

II. PENINSULAR DRAINAGE SYSTEM

➤ The peninsular river system can be categorised into the following sections:

- ✓ The East flowing rivers
- ✓ The West flowing rivers

WEST FLOWING RIVERS

- They form estuaries.
- Ex: Narmada, Tapi



EAST FLOWING RIVERS

- These rivers make deltas at their mouth.
- Ex: Ganga, Krishna, Kaveri





3. ORIGIN, NATURE & CHARACTERISTICS

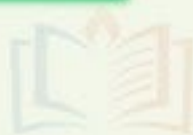
II. PENINSULAR DRAINAGE SYSTEM

SMALL RIVERS FLOWING TOWARDS THE WEST

- Several of these small rivers are important for the production of hydroelectricity because of steep slopes and waterfalls.
- ✓ **Gujarat:** The Shetrunji, The Bhadra, The Dhadhar, Sabarmati, Mahi
- ✓ **Maharashtra:** The Vaitarna, The Kalinadi, Bedti river, The Sharavati.
- ✓ **Goa:** Mandovi, Juari
- ✓ **Kerala:** Periyar, Bharathapuzha, The Pamba river.

SMALL RIVERS FLOWING TOWARDS THE EAST

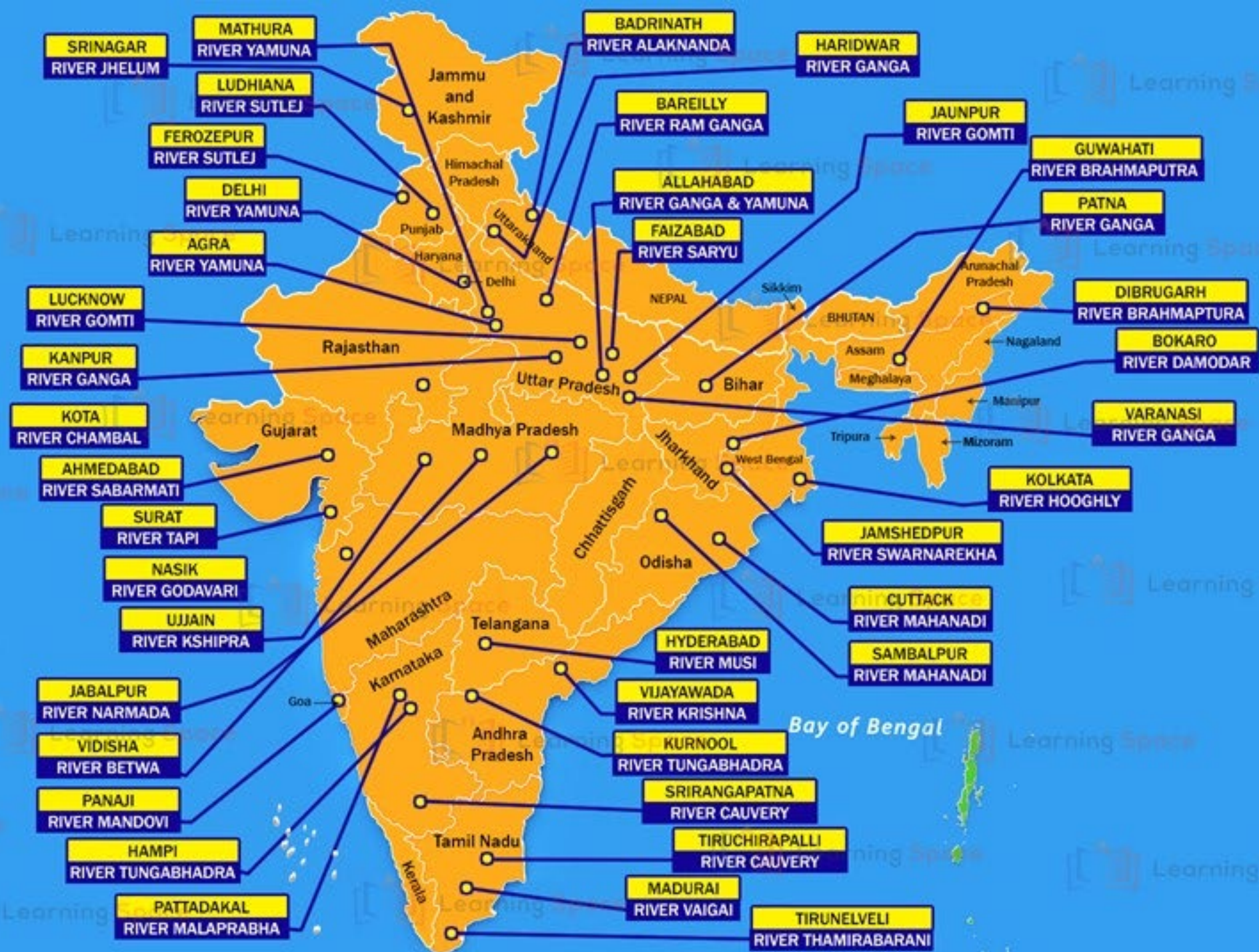
- These are small rivers which join the Bay of Bengal.
- The following are some of the rivers which are important in their own way.
 - ✓ The Subarnrekha
 - ✓ The Baitarni
 - ✓ The Brahmani
 - ✓ The Vamsadhara
 - ✓ The Penna
 - ✓ The Palar
 - ✓ The Vaigai



KNOW THE DIFFERENCES!

HIMALAYAN RIVERS	PENINSULAR RIVERS
Origin where Himalayan mountains are covered with glaciers.	Origin in Peninsular plateau and central highland.
Perennial; Receive water from glacier and rainfall.	Seasonal; Dependent on monsoon rainfall.
These are very large in size.	These are small basins.
These flow through steep sided V-shaped valleys.	These flow in comparatively shallow valleys.
These rivers flow across the young folded mountains and are still in a youthful stage.	These rivers have been flowing in one of the oldest plateaus and have most reached their base levels of erosion
When these rivers enter the plains, there is a sudden reduction in the speed of the flow of water. Under these circumstances these rivers form meanders and often shift their beds.	Hard Rock surfaces and non alluvial character of the plateau permits little scope for the formation of meanders.
These rivers make only deltas.	These rivers make deltas and estuaries.

FAMOUS CITIES & RIVER BANKS



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